Reasoning in Hinglish: Does Code-Mixed Chain-of-Thought Help Temporal Reasoning?

1 Question

Sarvam-105B gains far more from chain-of-thought (CoT) on English temporal questions than on Hindi ones. Hindi CoT even hurts on date subtraction and interval date. Hinglish, Hindi written in Roman script and mixed with English, is how many Indian users actually write. This experiment asks two questions:

- Does asking the model to reason in Hinglish close the English–Hindi CoT gap?
- What does reasoning in Hinglish do to English questions?

2 Setup

Prompt. The question is given in its own language (English or Hindi) with this instruction: *"Solve the following question by reasoning through it step by step. Your reasoning must be written in Hinglish, naturally mixing Hindi and English using only the Roman (Latin) script. Do not write in Devanagari script."* The answer-line format is the same as in the standard runs.

Data. The first 50 questions of every task and difficulty level of the Temporal Reasoning Dataset: 2,250 questions. Each is asked in English and Hindi, with no distractor, a time-related distractor and an unrelated distractor, which gives 13,500 Hinglish responses. The question text and gold answers are identical to the standard runs. Settings are also the same (Sarvam-105B, thinking off, temperature 0, 4,096 max tokens).

Comparison. The full-grid analysis pipeline is run twice on the same 2,250 questions:

- *Standard CoT*: the normal CoT responses, reasoning in the question's language.
- *Hinglish CoT*: the same responses with CoT replaced by Hinglish CoT in both languages.

Direct answers are shared. Intervals are 95% cluster bootstraps over questions ($B = 2000$). Task-level significance uses exact McNemar tests on paired responses, Holm-corrected over the 18 task–language tests.

3 Results

3.1 Hinglish CoT hurts English and does nothing for Hindi

	English questions				Hindi questions			
Task	Direct	Std	Hing.	Δ_{Hing}	Direct	Std	Hing.	Δ_{Hing}
Date addition	85.6	98.0	87.3	+1.7	82.8	85.1	85.9	+3.1
Date subtraction	86.0	95.2	70.7	**−15.3***	84.8	66.0	71.0	**−13.8***
Date duration	84.0	99.3	92.9	+8.9*	78.9	88.3	82.8	+3.8
Date recurrence	22.1	90.3	53.8	+31.7*	2.8	20.1	18.9	+16.1*
Day of week	14.9	69.5	34.7	+19.8*	15.2	21.0	23.5	+8.3*
Interval date	43.2	50.7	38.5	−4.7	46.8	23.9	23.6	**−23.2***
Time addition	70.5	95.9	82.9	+12.4*	75.6	73.5	72.3	−3.3
Time subtraction	64.7	97.7	85.3	+20.6*	64.9	75.2	74.2	+9.3*
Time duration	65.7	98.9	90.0	+24.3*	44.7	80.1	80.9	+36.3*
All tasks	59.7	88.9	70.9	+11.2*	55.2	59.4	59.4	+4.2*

Table 1: Accuracy (%) with direct answers, standard CoT (Std) and Hinglish CoT (Hing.) on the same 2,250 questions. Δ_{Hing} is Hinglish CoT minus direct, in points. * marks $p < 0.05$ (McNemar, Holm-corrected; uncorrected for the pooled row). Significant negative deltas are in bold.

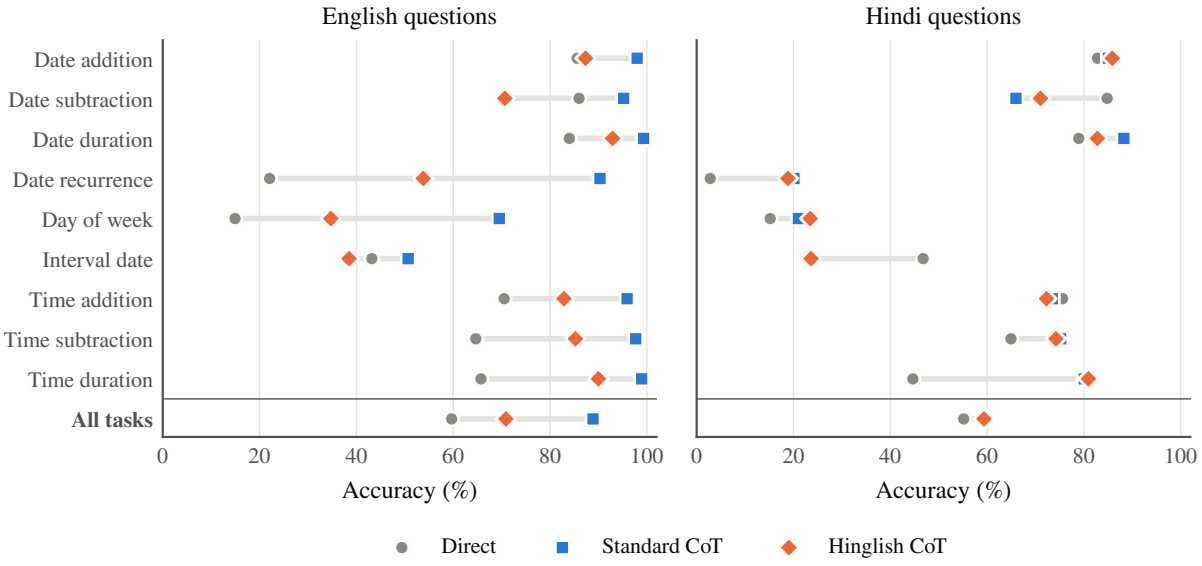


Figure 1: Accuracy per task. The gray bar spans the three conditions. For English questions, Hinglish CoT (orange) falls well short of standard CoT (blue) on every task. For Hindi questions, the two CoT conditions are close everywhere.

- **English questions:** CoT accuracy falls from 88.9% to 70.9%, and the gain over direct answers shrinks from +29.2 to +11.2 points. Hinglish CoT does worse than English CoT on every task. The largest losses are day of week (69.5 → 34.7), date recurrence (90.3 → 53.8) and date subtraction (95.2 → 70.7). Date subtraction becomes significantly *worse* than answering directly (−15.3). This is the failure Hindi CoT shows, now appearing in English.

- **Hindi questions:** accuracy is unchanged (59.4% in both). Every task-level change is within noise. Hinglish does not fix the two Hindi CoT failures: date subtraction (-13.8 vs. -18.8) and interval date (-23.2 vs. -22.9) are still significantly worse than direct.

3.2 The gap closes only because English gets worse

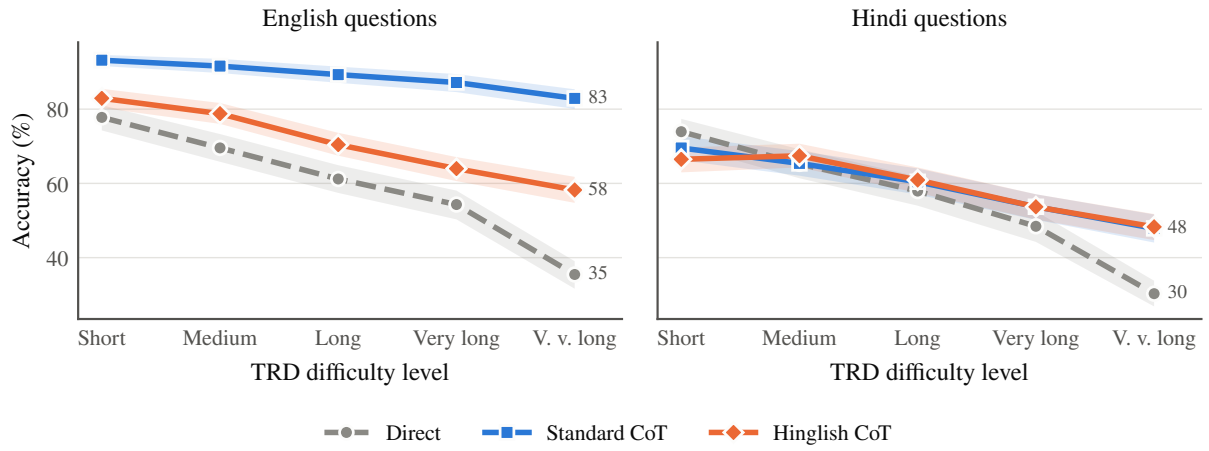


Figure 2: Accuracy over all nine tasks by difficulty level, with 95% cluster-bootstrap bands. For English questions, Hinglish CoT loses 10 points to English CoT at short and 25 points at very-very-long. For Hindi questions, Hinglish CoT and Hindi CoT coincide at every level.

	Standard CoT	Hinglish CoT
CoT gain, English questions	+29.2 [+27.5, +30.9]	+11.2 [+9.6, +12.8]
CoT gain, Hindi questions	+4.2 [+2.6, +5.9]	+4.2 [+2.5, +5.9]
EN−HI gap, five date tasks, short	+28.2 [+23.9, +32.8]	+17.5 [+13.1, +21.9]
EN−HI gap, five date tasks, very long	+40.6 [+35.7, +45.3]	+8.3 [+4.0, +12.3]
Median completion tokens, English Q.	353	233
Median completion tokens, Hindi Q.	212	223
Cut off before answering, English Q.	3.3%	0.8%
Cut off before answering, Hindi Q.	0.6%	0.4%
CoT distraction penalty, English, time-related	+0.4 [−1.9, +2.7]	+0.1 [−2.6, +2.9]
CoT distraction penalty, Hindi, time-related	+0.2 [−2.6, +2.9]	+0.8 [−1.9, +3.6]
CoT distraction penalty, English, unrelated	−1.9 [−4.2, +0.4]	−3.8 [−6.5, −1.1]
CoT distraction penalty, Hindi, unrelated	−0.3 [−3.0, +2.4]	+1.6 [−1.1, +4.3]

Table 2: Headline measures, standard vs. Hinglish CoT, with 95% cluster-bootstrap intervals in points. The EN−HI gap is English minus Hindi CoT accuracy on the five date tasks. The CoT distraction penalty is the accuracy drop from a distractor under CoT minus the drop under direct answering.

On the five date tasks, the English–Hindi CoT gap at very long falls from $+40.6$ to $+8.3$ points (Table 2). Figure 2 shows why: Hindi accuracy stays the same while English accuracy drops. This is not evidence that Hinglish helps Hindi. Hinglish traces for English questions are also shorter (median 233 vs. 353 tokens), and distractor penalties stay null in both languages. The one exception is English with an unrelated distractor (-3.8): Hinglish CoT was hurt slightly *less* by the distractor than direct answers were, which is most likely

chance.

3.3 The model ignores the script instruction on Hindi questions

Each Hinglish trace (excluding the answer line) is classified by script:

- *Devanagari mix*: at least 10% of the words are Devanagari.
- *Roman Hinglish*: no Devanagari, and at least 5% of the Latin words are common romanised Hindi function words (*hai, ko, mein, toh, nahi, …*).
- *Plain English*: everything else.

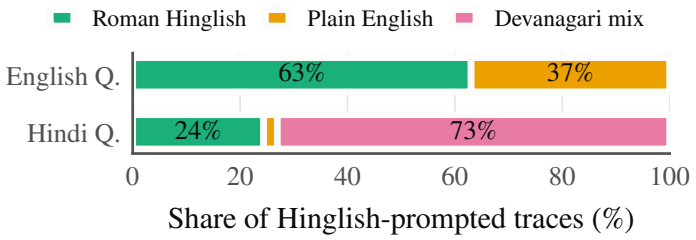


Figure 3: Script style of Hinglish-prompted traces, by question language.

For Hindi questions, 73% of traces are mostly Devanagari Hindi with English technical words, despite the explicit instruction. Only 25.5% contain no Devanagari at all, and the median Devanagari share is 68%. For English questions the model never uses Devanagari, but 37% of traces are essentially plain English.

> *Hindi question, "Hinglish" trace (Devanagari mix, correct)*
> सबसे पहले, हमें दिए गए दोनों time के बीच का difference निकालना है। शुरुआती time है 19:52 और final time है 20:17। … 8 + 17 = 25 minutes।

3.4 On English questions, more Hinglish means a larger loss

Question	Trace style	n	Hing.	Std	Hing. only	Std only
English	Roman Hinglish	4,109	68.0	89.3	138	1,013
English	Plain English	2,374	79.1	88.5	103	327
Hindi	Roman Hinglish	1,631	64.1	66.9	187	233
Hindi	Plain English	166	68.7	60.2	30	16
Hindi	Devanagari mix	4,882	57.8	57.1	732	696

Table 3: Each Hinglish trace compared with the standard CoT response to the same question and distractor condition. Hing. and Std are accuracies (%). The last two columns count pairs where only one of the two is correct. Styles with fewer than 100 traces are omitted.

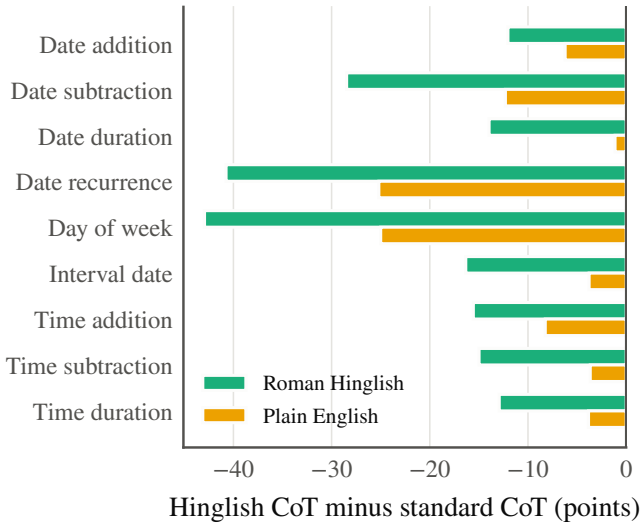


Figure 4: English questions: Hinglish CoT accuracy minus standard CoT accuracy on the same items, by task and trace style. Roman Hinglish traces lose more than plain-English traces on every task.

For English questions, Roman Hinglish traces score 68.0% against 89.3% for English CoT on the same items. In 1,013 pairs only English CoT was right; in 138 only Hinglish was. Plain-English traces under the Hinglish prompt lose less (79.1% vs. 88.5%). This ordering holds in all nine tasks (Figure 4). Standard CoT does about equally well on both groups (89.3% vs. 88.5%), so the Roman Hinglish group is not just the harder questions. For Hindi questions, Devanagari-mix traces match standard Hindi CoT (57.8% vs. 57.1%), as expected, since they are close to it.

Wrong Roman Hinglish traces show the arithmetic slips and shortcuts usually seen in Hindi CoT. Standard English CoT answered both of these questions correctly:

> *Date subtraction, short. "Today is 2025-05-01, what was the date 3 days ago?" Gold: 2025-04-28*
> Step 3: 2025-05-01 mein se 3 days minus karne padenge.
> Step 4: 01 minus 3 karne par hum 31 April par aayenge.
> Step 5: Isliye, 3 days ago ki date 2025-04-30 thi. **Answer: 2025-04-30**

> *Date recurrence, short. "Today is 2025-06-27, and I have a recurrence every 3 days. Without counting today, what is the date of the 1 occurrence?" Gold: 2025-06-30*
> Step 3: Agar aaj ko count nahi karenge, toh pehli recurrence kal se start hogi.
> Step 4: 3 days ka interval hai, toh kal 2025-06-28 hoga pehla occurrence. **Answer: 2025-06-28**

4 Takeaways

1. Hinglish CoT does not close the English–Hindi gap. Hindi accuracy is unchanged, and the gap shrinks only because English drops by 18 points.
2. Reasoning in Hinglish is costly on English questions, and the cost grows with difficulty and with how much Hindi the trace contains. Date subtraction and day of week break down the way they do in Hindi CoT. Together with the en-cot result (Hindi questions with English reasoning recover most of the gap), this suggests that reasoning in Hindi, in either script, is what limits Sarvam-105B's CoT, more than reading a Hindi question.

3. The model does not follow the Roman-script instruction on Hindi questions: 73% of its traces are mostly in Devanagari. So the Hindi arm mostly tests code-mixed Devanagari Hindi, not romanised Hinglish.

5 Caveats

- The trace-style thresholds and the function-word list are our own choices. The model picks the style itself, so the style comparison is descriptive, not randomised.
- Some measures from the full-grid pipeline do not carry over to Hinglish and are left out:
  - *Language drift* assumes Hindi traces should be in Devanagari.
  - *Verification* uses an English/Hindi phrase regex (the English rate falls from 86% to 15%, mostly because the regex misses Hinglish phrasing).
  - *Distractor mention*: Hindi distractors are in Devanagari, so romanised mentions are not matched.
- With 50 questions per task and level, per-task and per-level estimates are noisier than in the full grid.

Reproduction. The analysis is in `FINDINGS.md` of the folder `experiments/2026-10-31_hinglish-analysis-aayush/`, which lists the commands. This report's figures and tables come from `report/make_report.py`.